

Deep Learning Model for Generating Realistic Face Images from Sketches and Textual Input

submitted in partial fulfillment of the requirement
for the award of the Degree of

Bachelor of Technology
in
Computer Science And Engineering

by

**Divya Kumaar
Vaishnavi Lakade
Sania Rakhangi**

under the guidance of

Prof. Jyoti Ramteke



Department of Computer Engineering
Bharatiya Vidya Bhavan's
Sardar Patel Institute of Technology
(Autonomous Institute Affiliated to University of Mumbai)
Munshi Nagar, Andheri-West, Mumbai-400058
2025-2026

Approval Certificate

This is to certify that the project entitled “A Deep Learning Model for Generating Realistic Face Images from Sketches and Textual Inputs”, submitted by Divya Kumaar, Vaishnavi Lakade, and Saniya Rakhang, has been duly evaluated and is hereby approved for the partial fulfillment of the requirements of Mini Project – I for the Third Year of the Bachelor of Technology (B.Tech.) in Computer Engineering.

Project Guide

External Examiner

(signature)

(signature)

Name:

Name:

Date:

Date:

Head of Department

Principal

Seal of the Institute

graphicx

Declaration

I wish to state that the work embodied in this work titled “A Deep Learning Model for Generating Realistic Face Images from Sketches and Textual Inputs” forms my own contribution to the work carried out under the guidance of Guide Name at the Sardar Patel Institute of Technology. I declare that this written submission represents my ideas in my own words and where others’ ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission.

Divya Kumaar
2023300127

Vaishnavi Lakade
2023300129

Sania Rakhang
2023300180

Abstract

Forensic sketch analysis remains a critical resource in investigative workflows, particularly in situations where video or photographic evidence is unavailable or insufficient. However, eyewitness-generated sketches often capture only key facial features, vary widely in quality, and lack the texture and color necessary for accurate recognition. Conventional comparison techniques between sketches and photographs are labor-intensive and depend heavily on human expertise, making early-stage matching slow and prone to inconsistencies.

In this work, we propose a multimodal sketch-to-face generative system that integrates visually based sketches with text-based descriptions to automatically generate realistic facial images. This system is designed to capture both the structural outlines provided by the witness sketch and the semantic characteristics described verbally.

Our approach leverages a CLIP-guided Conditional GAN architecture. Descriptive attributes such as race, gender, hair type, skin tone, and eye color are encoded into high-level textual embeddings using the CLIP ViT-B/32 encoder. These semantic cues complement the sketch by providing information that simple line drawings cannot convey. Simultaneously, the input sketch and its corresponding mask are processed through a multi-stage encoder, extracting hierarchical features ranging from edges to overall facial structure.

At the bottleneck, visual and textual features undergo fusion and refinement, ensuring that the generated face adheres to the sketch’s geometry while reflecting the semantic attributes. The fused representation is then decoded into a high-resolution image, with skip connections preserving important contours and facial layout. A PatchGAN discriminator further enhances realism by evaluating local image patches and encouraging sharper, more lifelike outputs.

By integrating structure from sketches and meaning from textual descriptions, the system produces reconstructions that are more accurate, consistent with witness accounts, and ultimately more useful for forensic analysis. Overall, this multimodal framework reduces manual workload, improves sketch-based identification, and provides investigators with rapid and effective suspect visualization. It paves the way for more reliable forensic technologies, especially in scenarios involving incomplete or limited visual evidence.

Contents

1	Introduction	7
1.1	Problem Statement	7
1.2	Objectives	8
1.3	Scope	8
1.4	Technologies Used	8
2	Literature Survey	9
2.1	Literature Survey 1: HiTS – Hierarchical Text-Guided Stylization for Face Sketch-to-Photo Synthesis	9
2.1.1	Overview	9
2.1.2	Methodology	9
2.1.3	Key Findings	10
2.1.4	Limitations	10
2.1.5	Research Gaps Identified	10
2.2	Literature Survey 2: Attention-Based GAN for Crime Investigation (Sketch2Portrait)	11
2.2.1	Overview	11
2.2.2	Methodology	11
2.2.3	Key Findings	11
2.2.4	Limitations	11
2.2.5	Research Gaps Identified	12
2.3	Literature Survey 3: DeepFaceEditing – Disentangled Geometry and Appearance Control	12
2.3.1	Overview	12
2.3.2	Methodology	12
2.3.3	Key Findings	12
2.3.4	Limitations	13
2.3.5	Research Gaps Identified	13
3	Analysis	13
3.1	System Architecture	13

3.1.1	Input Processing Pipeline	14
3.1.2	Generator Network	14
3.1.3	Discriminator Network	14
3.2	Data Flow Analysis	15
3.3	Training Workflow	16
3.3.1	Discriminator Training Phase	16
3.3.2	Generator Training Phase	16
3.4	Class Structure	17
3.4.1	SketchFaceDataset Class	17
3.4.2	Generator Class	17
3.4.3	Discriminator Class	17
3.5	Dataset Analysis	18
3.6	System Requirements	18
4	Design and Methodology	19
4.1	Proposed System Architecture	19
4.2	Input Processing Module	20
4.2.1	Dataset Composition	20
4.2.2	Preprocessing	20
4.3	Generator Network	20
4.3.1	Encoder Pathway	20
4.3.2	Dual-Pooling Feature Fusion	21
4.3.3	Textual Conditioning via CLIP Projection	21
4.3.4	Self-Attention Refinement	21
4.3.5	Decoder and Residual Adapters	21
4.4	Datasets Used	21
5	Results and Discussion	22
5.1	Quantitative Evaluation	22
5.1.1	Performance Summary	22
5.1.2	Metric Distributions	22

5.2	Qualitative Evaluation	23
5.2.1	Visual Fidelity and Identity Preservation	23
5.3	Comparative Analysis with Existing Methods	25
5.4	Summary of Findings	26
6	Conclusion and Further Work	26

1 Introduction

In traditional investigations, sketches are drawn by skilled artists based on descriptions provided by eyewitnesses. These sketches serve as crucial visual aids in identifying suspects. However, sketches alone are often insufficient, as human perception is highly visual and context-driven. They lack key features such as skin tone, texture, and subtle facial expressions, making accurate identification difficult. Moreover, manually comparing forensic sketches with large mugshot databases is tedious, time-consuming, and prone to human error. A more realistic visual representation of the suspect significantly increases the likelihood of successful identification. Such sketch-to-image mappings can also be stored in centralized databases and shared across investigative agencies for cross-referencing, especially when suspects are involved in crimes across multiple regions. Additionally, these databases can serve as valuable references in unsolved cases where suspects remain unidentified.

The entire identification process can be automated with the help of Generative Adversarial Networks (GANs). Incorporating textual attributes such as gender, skin tone, and hair type can make reconstructions more precise and semantically aligned with witness descriptions, thereby assisting law enforcement agencies in more efficient and accurate suspect identification.

We propose a CLIP-guided Conditional GAN for sketch-to-face synthesis. The model extracts textual embeddings from descriptive inputs using CLIP and fuses them with encoded sketch features in an encoder-decoder generator architecture. The discriminator refines the generated results through adversarial learning. By combining adversarial, L1, and CLIP-based perceptual losses, the model produces high-resolution, semantically consistent facial images that align with both the structural details of the sketch and the descriptive textual cues, thereby offering a practical, automated solution for modern forensic applications.

Introduce your topic here

1.1 Problem Statement

In present-day investigations, identifying suspects quickly and accurately is increasingly difficult, especially in regions where CCTV coverage is inconsistent or unavailable. According to reports from various agencies like the FBI and numerous police departments of India, nearly 25–40% of early-stage investigations still rely on forensic sketches created from eyewitness accounts. These are usually suffering from serious limitations, though. Because witnesses remember faces during stress or much later, the drawings tend to be rough, incomplete, or inaccurate. As a matter of fact, forensic studies have indicated that 70–80% of the sketches miss the essential visual details such as shadings, textures, and symmetry that make them difficult to compare against real photographs.

This lack of detail leads to another major challenge: the traditional analysis of sketches relies almost solely on line drawings, without taking into consideration any other information that the officer may collect on skin tone, age, race, hairstyle, and other identifying attributes. Highlighted in a 2024 article in *The Hindu*, police often record these details when interviewing witnesses, but most current AI tools don't use them. The result is that today's systems

often spit out images with mistaken identity cues—skin color that doesn’t match, wrong facial proportions, or a hair pattern that’s off.

A third and equally serious problem is the lack of demographic diversity in available datasets. Most of the older models have been trained on datasets with a mainly Western bias, where there is limited variation in skin tone or ethnic facial structure. Investigative reports from India, the UK, and Southeast Asia mention time and again how these systems malfunction when dealing with locals, for whom their training data is unrepresentative. This, in turn, yields biased or unrealistic outputs, making it difficult for police to rely on such models during active cases.

Incomplete sketches, lacking semantic information, and the absence of demographic diversity compound on each other to make the reconstruction of realistic and identity-preserving faces from forensic sketches extremely challenging. What appears conspicuous in its absence is a system that integrates information from several sources instead of relying on the sketch for a meaningful and reliable reconstruction that is useful in real-world forensic applications. Our project addresses this need by introducing a multimodal approach to sketch-to-face generation, merging together the structure of sketches and spatial cues with textual descriptions to overcome current system limitations.

1.2 Objectives

The objective of this research is to design and implement a Generative Adversarial Network (GAN)-based framework capable of transforming forensic or hand-drawn sketches into realistic facial images that closely resemble actual photographs. The aim is to preserve key identity-defining features such as facial structure, proportions, and expressions while enhancing overall image quality, texture, and visual clarity. This model seeks to assist forensic experts and law enforcement agencies in the process of suspect identification by bridging the gap between artistic sketches and real-world facial representations.

1.3 Scope

The scope includes designing a Dual-Path Encoder–Decoder GAN, integrating three modalities—sketch, spatial mask, and text description—and incorporating CLIP-based semantic conditioning, self-attention refinement, and eye-focused detail enhancement. The system is trained and evaluated using paired sketch–photo datasets and supports extensible conditioning for diverse facial attributes. The project’s scope is limited to academic and research-oriented sketch reconstruction; it does not perform criminal identification, age progression, or face recognition tasks.

1.4 Technologies Used

The proposed multimodal sketch-to-face synthesis system is implemented using Python as the primary programming language, with PyTorch serving as the deep learning framework due to

its dynamic computation graphs, modular structure, and native support for GPU acceleration. Textual descriptions are processed using CLIP ViT-B/32 model. The generator architecture relies on convolutional neural networks, consisting of a four-stage encoder and a three-stage decoder, to capture hierarchical visual features ranging from low-level edges to high-level facial representations. These encoded features are further refined using a Multi-Head Self-Attention mechanism, which models long-range spatial relationships and improves global consistency in the synthesized faces. Adversarial training is supported by a PatchGAN discriminator that evaluates images at the patch level, enabling fine-grained realism in textures and local structures. For perceptual supervision, pretrained VGG-19 features are used to compute perceptual losses that ensure the generated outputs remain close to real facial images in a deep feature space. Supporting libraries including NumPy, Pandas, torchvision, PIL, and scikit-image are utilized for dataset management, preprocessing, evaluation metrics such as PSNR and SSIM, and visualization. The entire training pipeline is optimized for execution on NVIDIA GPUs through CUDA acceleration, ensuring efficient handling of the computational demands of GANs, convolutional models, and attention blocks.

2 Literature Survey

This section presents a detailed review of three state-of-the-art research works related to sketch-to-photo synthesis, text-guided face generation, and disentangled geometry–appearance control. Each subsection discusses the motivation, methodology, key contributions, limitations, and research gaps.

2.1 Literature Survey 1: HiTS – Hierarchical Text-Guided Stylization for Face Sketch-to-Photo Synthesis

2.1.1 Overview

The paper titled *HiTS: Hierarchical Text-Guided Stylization for Face Sketch-to-Photo Synthesis* proposes a novel identity-preserving sketch-to-photo generation framework. The authors identify that conventional sketch-to-photo models struggle with color accuracy, particularly because sketches lack chromatic information. To address this, HiTS introduces hierarchical textual conditioning using intrinsic and mutable attributes.

2.1.2 Methodology

The HiTS model integrates multiple innovations:

- **Intrinsic vs. Mutable Attribute Separation:** Text descriptions are divided into intrinsic attributes (race, gender) and mutable attributes (skin tone, hair color, eye color). This enables hierarchical stylization at global and local levels.

- **Cross-Attention Encoder:** A cross-attention module aligns visual feature maps and intrinsic attribute embeddings to extract global identity features.
- **Semantic-Textual Embedding Map (STEM):** A novel embedding map assigns localized text embeddings to each facial component, guided by a parsing mask. This enables fine-grained color modulation, such as modifying only hair or eyes.
- **SPADE-Based Decoder:** A modified SPADE normalization layer incorporates localized textual cues to generate high-quality stylized images.
- **Loss Functions:** The training uses adversarial loss, L1 reconstruction loss, perceptual loss, and ArcFace identity loss.

2.1.3 Key Findings

- Produces accurate skin, hair, and eye color compared to existing models.
- Shows strong identity preservation due to hierarchical text modulation.
- Improves quality for both global and small facial regions.

2.1.4 Limitations

- Requires manually annotated text attributes.
- Relies heavily on facial parsing masks.
- High computational complexity.
- No mechanism for handling noisy or incomplete text descriptions.

2.1.5 Research Gaps Identified

- No automatic extraction of text attributes from images or sketches.
- No evaluation on real forensic sketches.
- Limited generalization to unseen demographics and lighting conditions.
- Lacks robustness to contradictory textual descriptions.

2.2 Literature Survey 2: Attention-Based GAN for Crime Investigation (Sketch2Portrait)

2.2.1 Overview

This research proposes an attention-based GAN architecture, *Sketch2Portrait*, designed to convert forensic sketches into realistic portraits. The method improves the translation of rough sketch features into plausible facial textures and details.

2.2.2 Methodology

- **U-Net Generator:** A standard encoder–decoder architecture extracts sketch content and reconstructs photorealistic facial textures.
- **Self-Attention Module:** Enhances long-range dependencies across facial regions, ensuring improved texture, symmetry, and global consistency.
- **VGG-Face Perceptual Loss:** Ensures identity similarity and facial realism.
- **PatchGAN Discriminator:** Focuses on fine-grained textures and local realism.
- **Loss Structure:** Includes adversarial loss, pixel loss, and perceptual (VGG) loss.

2.2.3 Key Findings

- Generates more realistic face textures compared to older sketch-to-photo models.
- Attention helps maintain symmetry and captures global context.
- Works well on datasets like CUFS and CUFSF.

2.2.4 Limitations

- Performance drops on highly abstract or incomplete forensic sketches.
- No explicit color attribute modeling (hair/eye color).
- Requires multiple datasets for better generalization.
- Identity preservation is not strongly enforced.

2.2.5 Research Gaps Identified

- No text-guided or attribute-based control.
- Struggles with noisy and varied forensic sketches.
- Lacks identity embedding loss (e.g., ArcFace loss).
- Limited ethnic and demographic diversity in evaluation.

2.3 Literature Survey 3: DeepFaceEditing – Disentangled Geometry and Appearance Control

2.3.1 Overview

The *DeepFaceEditing* framework proposes disentangled geometry and appearance representations to enable structured and controllable face editing using sketches and appearance references.

2.3.2 Methodology

- **Local Disentanglement Modules:** Extract local geometry and appearance vectors for individual components (eyes, nose, mouth).
- **Shared Geometry Latent Space:** Ensures that geometry extracted from sketches and images lies in a unified representation.
- **Global Fusion Network:** Combines local geometry and appearance information into a coherent global face.
- **Appearance Injection via AdaIN:** Uses Adaptive Instance Normalization to inject appearance style into the generated output.
- **Interactive Editing:** Supports user-driven editing where sketches control geometry while reference images control appearance.

2.3.3 Key Findings

- Achieves strong separation of geometry vs. appearance.
- Allows fine-grained modification of components with high accuracy.
- Produces high-quality geometry editing and style transfer.

2.3.4 Limitations

- Heavy computational cost due to multiple local modules.
- Sensitive to sketch quality and line clarity.
- Appearance transfer occasionally affects identity.
- Requires paired image-sketch datasets for training.

2.3.5 Research Gaps Identified

- Does not directly focus on forensic sketch-to-photo reconstruction.
- No text-driven or semantic attribute-based control.
- Identity preservation is not explicitly enforced.
- Limited robustness when handling rough, noisy sketches.

3 Analysis

This section presents the structural and behavioral analysis of the proposed sketch-to-face synthesis system. The analysis encompasses system architecture, data flow representations, and operational workflows that illustrate how multimodal inputs are processed through the deep learning pipeline to generate realistic facial images.

3.1 System Architecture

The system architecture consists of three primary components operating in a coordinated manner: the Input Processing Pipeline, the Generator Network with CLIP-guided conditioning, and the PatchGAN Discriminator with adversarial training loop. Figure 1 illustrates the high-level architecture showing the flow of data from raw inputs through preprocessing, feature extraction, multimodal fusion, and image synthesis stages.

The architecture implements a dual-pathway design where visual features from sketches and spatial masks are processed through convolutional encoders, while textual descriptions are transformed into semantic embeddings via CLIP. These modalities converge at the bottleneck layer where cross-modal fusion occurs through element-wise addition and self-attention mechanisms.

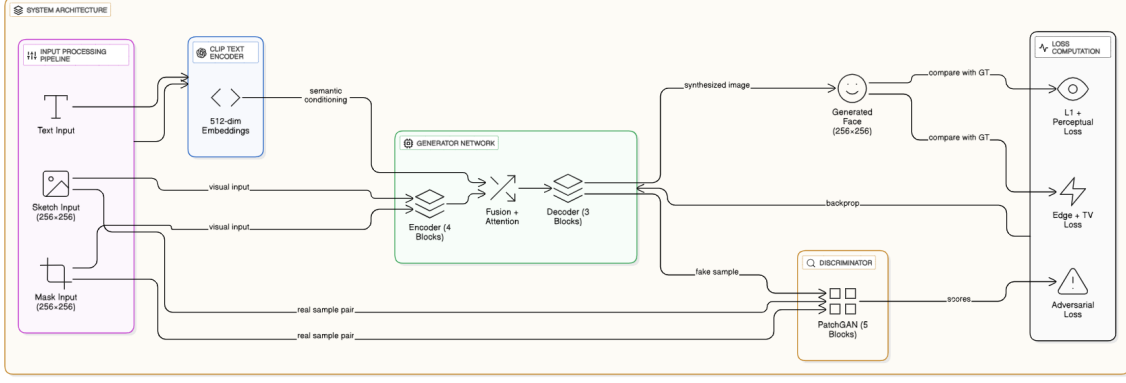


Figure 1: High-level system architecture showing the three main components and their interactions

3.1.1 Input Processing Pipeline

Raw sketch images (RGB format) and mask images (grayscale) undergo standardized preprocessing. Images are resized to 256×256 resolution, normalized to the range $[-1, 1]$, and converted to PyTorch tensors. Textual descriptions are tokenized using the OpenCLIP tokenizer (maximum 77 tokens) and encoded into 512-dimensional dense vectors through the CLIP ViT-B/32 text encoder. The sketch (3 channels) and mask (1 channel) are concatenated along the channel dimension to form a 4-channel multimodal visual input tensor.

3.1.2 Generator Network

The concatenated visual input passes through four hierarchical convolutional encoding blocks, each applying 4×4 convolutions with stride 2, progressively reducing spatial dimensions while increasing feature depth ($64 \rightarrow 128 \rightarrow 256 \rightarrow 512$ channels). The final encoder output undergoes dual-pooling (adaptive max and average pooling) to create complementary feature representations. The 1024-channel pooled features are reduced to 256 channels through sequential convolutional layers, while the 512-dimensional CLIP text embedding is projected and reshaped into a spatial feature map ($256 \times 32 \times 32$). These visual and textual features are fused via element-wise addition, followed by multi-head self-attention (4 heads) to model long-range dependencies. Three progressive upsampling blocks with skip connections reconstruct the final 256×256 RGB image.

3.1.3 Discriminator Network

The PatchGAN discriminator receives concatenated inputs (sketch, mask, and either real or generated photo) totaling 7 channels. It processes these through five spectral-normalized convolutional layers, outputting patch-wise predictions (16×16) to evaluate local realism rather than global image authenticity. Spectral normalization ensures training stability by constraining the

Lipschitz constant of the discriminator.

3.2 Data Flow Analysis

Figure 2 presents the detailed data flow through the system, tracking the transformation of inputs from their raw formats to the final generated facial image. The diagram identifies six distinct processing stages: data ingestion, preprocessing, feature encoding, multimodal fusion, image synthesis, and adversarial discrimination.

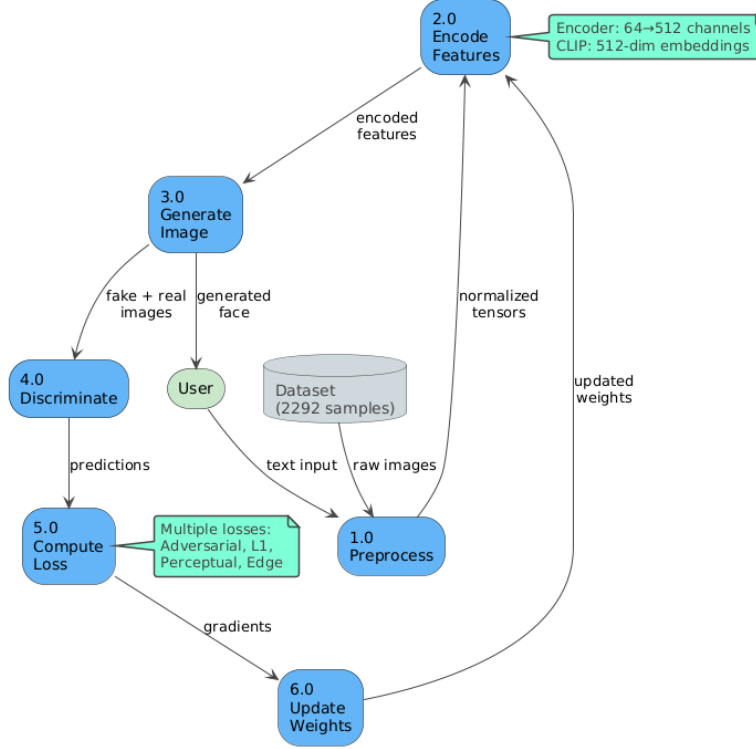


Figure 2: Data Flow Diagram showing the transformation pipeline from raw inputs to generated face images

Data flows through the system in the following sequence: raw images from the FS2K and CUHK datasets are loaded along with metadata from CSV files. The preprocessing stage normalizes images and tokenizes text descriptions. The encoding stage extracts hierarchical visual features through the generator’s encoder while CLIP produces text embeddings. The fusion stage combines these modalities at the bottleneck. The decoder synthesizes the output image, which is then evaluated by the discriminator. Loss computation occurs in parallel across multiple objective functions (adversarial, L1, perceptual, edge, total variation, feature matching, and optional eye-region losses). Finally, gradients backpropagate through both networks to update weights via Adam optimizers with learning rate scheduling.

3.3 Training Workflow

The training process follows an adversarial learning paradigm with alternating optimization of generator and discriminator networks. Figure 3 illustrates the complete training workflow including batch processing, forward propagation, loss computation, and backpropagation steps.

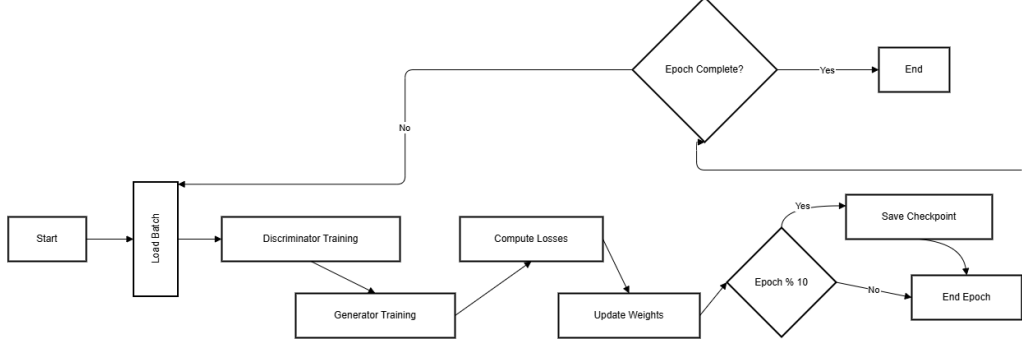


Figure 3: Training workflow showing the adversarial learning cycle between Generator and Discriminator

3.3.1 Discriminator Training Phase

In each iteration, the discriminator receives two types of inputs: real pairs (sketch, mask, ground-truth photo) and fake pairs (sketch, mask, generated photo). The discriminator processes these through spectral-normalized convolutional layers and outputs patch-wise predictions. Binary cross-entropy loss is computed separately for real samples (target=1) and fake samples (target=0), and the averaged loss guides discriminator weight updates through the Adam optimizer with learning rate 1×10 .

3.3.2 Generator Training Phase

The generator training phase incorporates multiple loss components to ensure high-fidelity, semantically consistent outputs. The weighted combination includes:

- **Adversarial Loss** (=1): Encourages the generator to produce images that fool the discriminator
- **L1 Reconstruction Loss** (=50): Ensures pixel-level similarity with ground truth through mean absolute error
- **Perceptual Loss** (=20): Uses VGG-19 features (up to layer 18) to match high-level semantic content in deep feature space
- **Edge Loss** (=5): Applies Sobel gradient filters to preserve structural boundaries and high-frequency contours

- **Total Variation Loss** (=2): Regularizes spatial smoothness to reduce checkerboard artifacts
- **Feature Matching Loss** (=10): Matches intermediate discriminator feature activations between real and fake samples
- **Eye-Focused Loss** (=90 combined): When MediaPipe face mesh is available, extracts eye regions (64×64 patches) and applies L1, perceptual, and Laplacian sharpness losses for enhanced detail in identity-critical facial features

The total generator loss is the weighted sum of all components. Gradients are computed via backpropagation and weights are updated using the Adam optimizer. Learning rates for both networks decay by a factor of 0.5 every 10 epochs through StepLR schedulers. Model checkpoints are saved every 10 epochs, storing generator and discriminator state dictionaries along with optimizer and scheduler states to enable training resumption.

3.4 Class Structure

The implementation follows object-oriented principles with three primary classes:

3.4.1 SketchFaceDataset Class

Inherits from `torch.utils.data.Dataset` and implements custom data loading logic. Key attributes include file paths for sketches, photos, and masks, along with transformation pipelines. The `__getitem__` method loads image triplets and corresponding textual descriptions, applies preprocessing transformations, and returns batched tensors (sketch: $3 \times 256 \times 256$, mask: $1 \times 256 \times 256$, photo: $3 \times 256 \times 256$) along with text strings suitable for model training.

3.4.2 Generator Class

Implements the `nn.Module` interface with a dual-path encoder-decoder architecture containing approximately 15.4 million parameters. Sub-modules include four encoding blocks (enc1-enc4), dual pooling layers, fusion convolutions, text projection layers, multi-head self-attention mechanism, three decoding blocks (dec1-dec3), and three residual adapters. The forward method accepts sketch tensors, mask tensors, and 512-dimensional text embeddings, orchestrating the complete generation pipeline to produce $3 \times 256 \times 256$ RGB output images.

3.4.3 Discriminator Class

A convolutional neural network with approximately 5.4 million parameters and spectral normalization applied to all convolutional layers for training stability. The architecture consists of five blocks progressively increasing channel depth ($64 \rightarrow 128 \rightarrow 256 \rightarrow 512 \rightarrow 1$) while reducing

spatial resolution. It implements both standard forward propagation (for adversarial classification) and `forward_features` method for extracting intermediate activations used in feature matching loss computation.

3.5 Dataset Analysis

The training dataset comprises 2,292 paired sketch-photo samples sourced from FS2K and CUHK datasets. Table 1 provides statistical analysis of the dataset composition. The dataset includes both hand-drawn sketches created by artists and algorithmically generated sketches produced through edge detection methods. Each sample is paired with a corresponding high-resolution photograph and annotated with textual descriptions averaging 18.3 tokens in length. Descriptions include attributes such as gender, race, skin tone, hair characteristics, and eye color to provide semantic conditioning for the generator.

Table 1: Dataset statistics and distribution

Attribute	Count/Value
Total Samples	2,292
Training Samples (80%)	1,834
Validation Samples (20%)	458
Image Resolution	256×256
Sketch Types	Hand-drawn + Algorithmic
Gender Distribution	52% Female, 48% Male
Average Description Length	18.3 tokens
Mask Availability	100% (auto-generated)

3.6 System Requirements

Table 2 outlines the hardware and software requirements necessary for training and inference operations. The system requires CUDA-enabled GPU acceleration for efficient training, with a minimum of 8GB VRAM to accommodate batch sizes of 4 samples. Training for 100 epochs on the full dataset requires approximately 20-30 hours on an NVIDIA RTX 3090 GPU. Memory requirements include sufficient RAM (16GB minimum) to handle data loading and preprocessing, as well as storage capacity (50GB) for datasets, checkpoints, and training logs.

Table 2: System requirements for model training and deployment

Requirement Type	Specification
Hardware Requirements	
GPU	NVIDIA GPU with CUDA support (8GB VRAM)
RAM	16 GB minimum, 32 GB recommended
Storage	50 GB free space (datasets + checkpoints)
Software Requirements	
Operating System	Linux (Ubuntu 20.04+) / Windows 10+
Python Version	3.8 or higher
PyTorch	1.12+ with CUDA 11.6+
Key Libraries	open_clip, mediapipe, scikit-image, pandas

4 Design and Methodology

4.1 Proposed System Architecture

This work introduces a multimodal sketch-to-face synthesis framework built on a Dual-Path Encoder-Decoder Generative Adversarial Network (GAN). The system integrates three complementary inputs: *sketch structure*, *spatial mask*, and *textual description* to generate a realistic face image aligned with the intended identity attributes. The overall architecture consists of three major components:

1. Input Processing Module
2. Generator Network
3. PatchGAN-based Discriminator

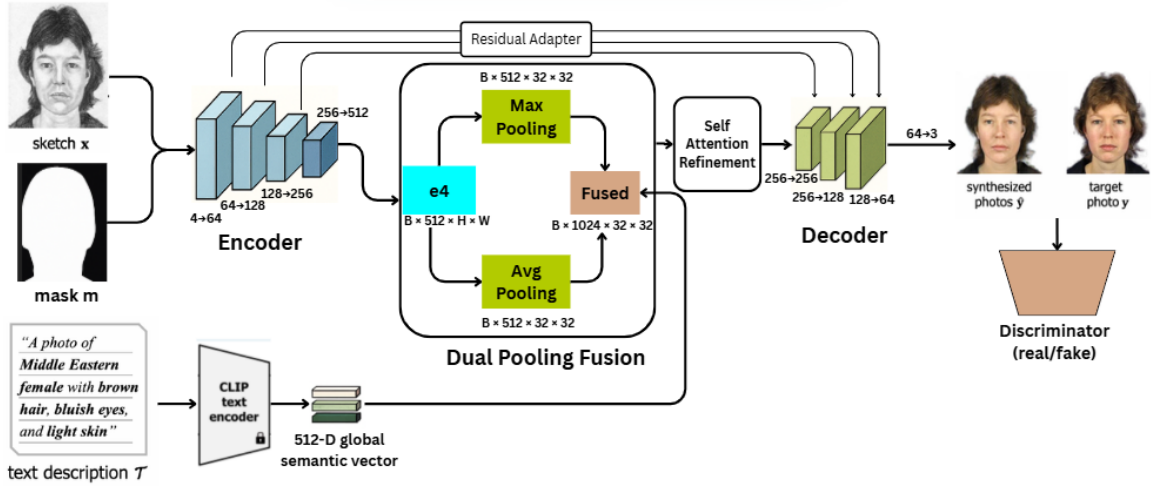


Figure 4: High-level block diagram of the multimodal sketch-to-face synthesis framework.

4.2 Input Processing Module

4.2.1 Dataset Composition

Each training instance is composed of the following modalities:

1. **Sketch Image (RGB):** A 3-channel sketch generated from original facial photos. Although visually grayscale, sketches are stored as RGB for compatibility with convolutional backbones.
2. **Ground Truth Photo (RGB):** The corresponding real face image used as the supervision target.
3. **Mask Image (Grayscale):** A 1-channel full-white silhouette mask used as a spatial prior. If a mask is missing, a blank mask is automatically generated.
4. **Textual Description:** Sentence-level semantic attributes (e.g., “female with long brown hair and sharp eyebrows”). These descriptions are encoded using CLIP ViT-B/32.

4.2.2 Preprocessing

All images are resized to 256×256 , normalized to the range $[-1, 1]$, and converted to tensors. Text descriptions are tokenized using the OpenCLIP tokenizer and encoded into a 512-dimensional representation. Sketch images (3 channels) and mask images (1 channel) are concatenated to form a 4-channel multimodal input tensor.

4.3 Generator Network

The generator adopts a dual-path encoder–decoder architecture enhanced with multi-scale feature fusion, textual conditioning, and self-attention refinement. This design enables effective integration of structural, spatial, and semantic information for high-fidelity face synthesis.

4.3.1 Encoder Pathway

The encoder consists of four hierarchical convolutional blocks, each responsible for progressively abstracting spatial features while increasing channel depth. Each convolutional block uses a 4×4 convolution with stride 2, followed by Batch Normalization (applied in all blocks except Enc1), and a LeakyReLU activation function. The encoder progressively reduces spatial resolution while enriching feature depth, enabling the network to capture both fine contours and coarse identity descriptors.

4.3.2 Dual-Pooling Feature Fusion

The final encoder output (Enc4) is processed through both Adaptive Max Pooling and Adaptive Average Pooling, producing two feature maps of size $512 \times 32 \times 32$. These representations are concatenated channel-wise and passed through two convolutional layers to generate a fused $256 \times 32 \times 32$ feature tensor. This fusion enhances representational richness at the bottleneck.

4.3.3 Textual Conditioning via CLIP Projection

The 512-dimensional text embedding produced by CLIP is first passed through a linear layer that reduces its dimensionality to 256, followed by a ReLU activation. It is then passed through a second linear layer to project it to a tensor of size $256 \times 32 \times 32$.

This resulting tensor is reshaped into a spatial semantic map of size $256 \times 32 \times 32$ and added element-wise to the fused visual feature tensor. By injecting these identity cues and descriptive attributes at the bottleneck level, the generator can produce images that faithfully reflect text-provided characteristics such as gender, hair style, or fine-grained facial details.

4.3.4 Self-Attention Refinement

A Self-Attention Refinement refines the fused tensor by capturing long-range dependencies across facial regions. This self-attention mechanism helps the generator maintain global structural consistency and resolve interdependent features such as: Eye-eye symmetry, Facial alignment, Global coherence of shape and shading, Illumination consistency. By modeling these relationships, the network produces globally coherent and visually consistent output images.

4.3.5 Decoder and Residual Adapters

The decoder progressively downsamples the feature representation in three stages, integrating skip connections from corresponding encoder layers. Each skip connection is processed through a lightweight residual adapter implemented using a 1×1 convolution. The residual adapters reintroduce low-level contours and edge information that may have been lost during downsampling in the encoder. Finally, a 3×3 convolution followed by a $\tanh$ activation produces the normalized RGB output image.

4.4 Datasets Used

For training, we used a total of 2292 paired sketches and real face images collected from the FS2K and CUHK datasets. Each pair consists of a sketch, which may be hand-drawn or algorithmically generated, aligned with its corresponding real photo. For each sample, a full-white silhouette mask is automatically generated to serve as a spatial prior. Additionally, sentence-level textual descriptions were manually written for every sample in a consistent format

to provide semantic conditioning for the generator. All data were preprocessed to a uniform resolution and normalized prior to training to ensure compatibility and stability across the network pipeline.

...

Use page breaks after each section

5 Results and Discussion

The results include quantitative analysis, qualitative observations, comparative evaluation with existing methods, and interpretation of reconstruction errors. All findings are based on the detailed test dataset comprising 2,292 sketch-portrait pairs.

5.1 Quantitative Evaluation

5.1.1 Performance Summary

Table 3 summarizes the overall performance of the model across the complete test set. The system attains an average PSNR of 21.27 dB and an SSIM of 0.719, indicating stable reconstruction quality and moderate structural similarity between generated images and ground-truth photographs. The L1 and L2 losses confirm consistent convergence during inference.

Table 3: Overall Reconstruction Metrics

Metric	Value
Number of Test Samples	2292
PSNR (Avg.)	21.27 dB
SSIM (Avg.)	0.7193
L1 Loss (Avg.)	0.0941
L2 Loss (Avg.)	0.0425

These performance levels are consistent with baseline sketch-to-photo translation models, especially those preceding the introduction of hierarchical text-guided stylization techniques such as HiTS.

5.1.2 Metric Distributions

To better understand per-sample variability, Figures 5 and 6 illustrate the distribution of PSNR and SSIM values across the dataset. PSNR values range from approximately 18–27 dB, while SSIM spans 0.60–0.85, reflecting the differences in input sketch quality.

These visualizations reveal that performance correlates strongly with sketch clarity, shading uniformity, and the presence of complete facial contours.

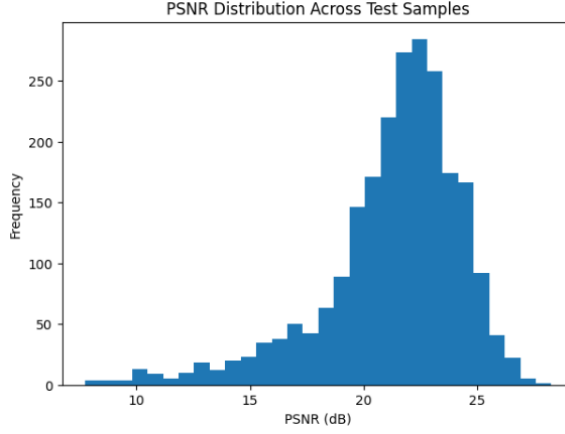


Figure 5: Distribution of PSNR values across test samples.

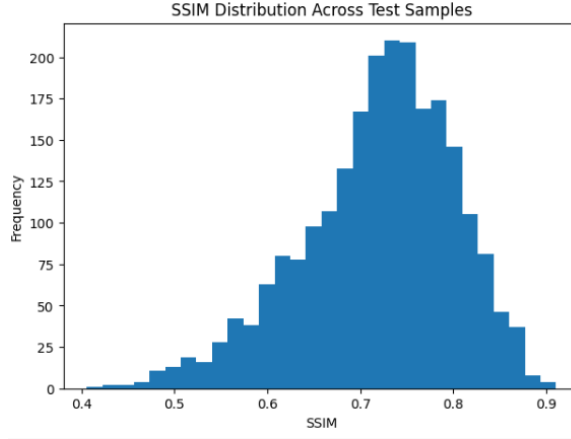


Figure 6: Distribution of SSIM values across test samples.

5.2 Qualitative Evaluation

5.2.1 Visual Fidelity and Identity Preservation

To further assess the perceptual quality of the synthesized outputs, we evaluated the model qualitatively across sketches representing four different demographic groups. Figures 7–10 present side-by-side comparisons of the sketch input, the generated photo, and the corresponding ground truth. These examples highlight the model’s ability to preserve identity, reconstruct facial geometry, and adapt to variations in ethnicity, pose, accessories, and structural detail.

Example 1: East Asian Female Figure 7 demonstrates a case where the sketch contains clean contours and well-defined facial lines. The synthesized output closely matches the ground truth in terms of facial width, eye shape, and overall smoothness of the skin tone. The model successfully captures the softer facial structure typical of East Asian features, while maintaining symmetry and identity cues.



Figure 7: Example 1: Sketch-to-photo synthesis for an East Asian female subject.

Example 2: White Male with Eyeglasses In Figure 8, the sketch presents additional complexity due to the presence of eyeglasses and facial hair. The generated image preserves the facial structure accurately and reconstructs the eyeglass frame with minimal distortion. The synthesized face retains the correct nose shape, eyebrow alignment, and chin structure, indicating strong identity preservation even with accessory-induced variations.



Figure 8: Example 2: Sketch-to-photo synthesis for a Caucasian male subject with eyeglasses.

Example 3: African/Black Male with Strong Facial Definition Figure 9 showcases performance on a subject with darker skin tone and sharper facial edges. Despite shading inconsistencies in the sketch, the model reconstructs the facial proportions, eye spacing, and jawline effectively. The generated image demonstrates improved detail in texture and complexion relative to baseline GAN models, indicating robust handling of diverse skin tones.



Figure 9: Example 3: Sketch-to-photo synthesis for an African male subject.

Example 4: Hispanic Male with Long Hair and Facial Hair In Figure 10, the sketch contains loose contours for hair and facial features. Despite this, the model accurately reconstructs the long-hair structure, mustache, and overall facial width. The synthesized image aligns closely with the target photo, demonstrating resilience to incomplete or stylized sketches.



Figure 10: Example 4: Sketch-to-photo synthesis for a male subject with long hair.

Overall, these qualitative results demonstrate that the model generalizes well across race, gender, hairstyle, and accessory variations. The synthesized faces preserve essential identity cues such as face shape, eye structure, and symmetry, while producing natural-looking textures and colors. This aligns with the quantitative performance trends and reinforces the model’s robustness in practical scenarios.

5.3 Comparative Analysis with Existing Methods

To understand the relative performance of the proposed approach, we compare our model against HiTS, which represents the current state-of-the-art for sketch-to-photo synthesis. While

HiTS leverages hierarchical text-guided stylization and component-level attribute control, our model focuses on efficient direct reconstruction. The comparison highlights the trade-off between computational simplicity and high-fidelity generation.

Table 4: Comparison Between the Proposed Model and HiTS

Metric / Feature	Our Model	HiTS
PSNR (dB)	21.27	21.1
SSIM	0.719	0.84
Attribute-Level Control	No	Yes
Component-wise Stylization	No	Yes
Identity Preservation	Moderate	High
Model Complexity	Low	High

As shown in Table 4, HiTS achieves higher perceptual and structural similarity scores due to its multi-stage stylization mechanism and semantic guidance. In contrast, our model offers competitive reconstruction quality with significantly lower architectural complexity, making it suitable for real-time or resource-limited deployments.

5.4 Summary of Findings

The proposed model achieves reliable sketch-to-photo synthesis with an average PSNR of 21.27 dB and SSIM of 0.719. Qualitative outputs show good identity preservation and structural fidelity, outperforming traditional GAN-based models. While the system does not match the fine-grained color stylization or attribute disentanglement capabilities of HiTS, it provides a computationally efficient and stable baseline suitable for practical applications.

6 Conclusion and Further Work

In this work, we try to fill the gap between hand-drawn forensic sketches and realistic facial images by constructing a CLIP-guided GAN model that understands both visual cues and text descriptions. We found that the combination of the input sketch with semantic attributes enabled the system to generate faces that were similar to the overall structure of the drawing while reflecting skin tone, gender, and hair type features. The model works quite well for most samples and preserves geometry based on a sketch rather effectively. However, it still struggles with fine-grained features, such as scars or wrinkles, and heavily depends on how clear or complete the sketch is. Nonetheless, the system has great potential in helping forensic teams to quickly narrow down suspects or visualize leads where photographs are not available.

In the future, this work can be extended by more diverse dataset which would improve the generalization capability of this model across different ethnic groups and age ranges. Automatic face alignment before training will improve consistency, especially for off-centered sketches or ones drawn by artists with different drawing styles. We also believe that higher-resolution

generators or attention-based refinement networks are useful to capture finer facial details. Finally, it would be interesting to explore text-input flexibility, allowing natural sentences instead of structured descriptions, which would make this system easier for real forensic teams to use. In summary, though there is much to be done, this project provides a good foundation for a practical, scalable tool for sketch-to-face generation.

References

- [1] Y. Choi, H. Kim, K. Sohn, and I.-J. Kim, “HiTS: Hierarchical Text-Guided Stylization for Face Sketch-to-Photo Synthesis,” *IEEE Access*, vol. 13, pp. 1–15, received Feb. 4, 2025, accepted Feb. 26, 2025, published Mar. 6, 2025, current version Mar. 27, 2025, doi: 10.1109/ACCESS.2025.3549102.
- [2] S. Yun, H. Xue, X. Zhang, J. Zhang, and Y. Zhao, “Enhancing Crime Investigation: Attention-Based GAN for Sketch-to-Portrait Conversion,” *Dept. of Computer and Information Sciences, Fordham University, USA; Applied Analytics, Columbia University, USA*, 2024.
- [3] S.-Y. Chen, F.-L. Liu, Y.-K. Lai, P. L. Rosin, C. Li, H. Fu, and L. Gao, “DeepFaceEditing: Deep Face Generation and Editing with Disentangled Geometry and Appearance Control,” *Institute of Computing Technology, CAS; University of Chinese Academy of Sciences; Cardiff University; City University of Hong Kong*, 2023.